Healthy Skepticism · Session 4 take-home summary

Fault 5: the causal leap

Two things move together. When one goes up, the other tends to go up with it. Plot them over time and the lines echo each other.

That is a fact about the data, and it is where every claim in this session begins. It is not an explanation, and it never says why. Treating the pattern as though it had named a cause is Fault 5.

Three explanations for a correlation

Structure	What it says
Pure coincidence	Nothing connects them.
Confounders	Something else produces both.
A genuine cause	The claim is true.

One notation runs through the whole session. A solid arrow means causes. A dashed line means moves together, which is all the data ever shows you. A crossed-out arrow is a cause that was looked for and not found. One box with arrows into two others is a single cause behind both.

> Only the third one is the claim being made.
> All three produce the same correlation, and the data cannot tell you which one you are looking at.

Coincidence: correlation without a cause

Tyler Vigen's Spurious Correlations project hunts for variables that happen to move together. Thousands of variables, every pair tried, hundreds of millions of pairs, and the tightest fits are kept. Search that many pairs and a near-perfect match turns up somewhere, meaning nothing.

The two charts below are convincing on their face. Neither has a cause behind it.

Saturn and the physical sciences

How far Saturn sits from the moon tracks how many physical sciences degrees America awards. $r = 0.987$, and $p = 1.2 \times 10^{-7}$, past five sigma. It would clear any journal's bar. It means nothing at all.

Compiled by Tyler Vigen, tylervigen.com, spurious correlation #2,656, CC BY 4.0. Planetary distances from Astropy; degrees from the NCES Digest of Education Statistics 2022. Over 2012-2021.

Cheese and BlackRock

American cheese eaten per person tracks BlackRock's share price, at $r = 0.94$ with $p = 8 \times 10^{-10}$. Both went up. That is the entire connection.

There is a check for this, and it is one you can follow without any statistics. If cheese moved the price, the years cheese rose fastest should be the years the price rose fastest. Correlate the year-to-year changes instead of the totals and it is gone: $r = 0.29$, $p = 0.23$.

> Any two things that grow will correlate. The trend is not the evidence.

Compiled by Tyler Vigen, tylervigen.com, spurious correlation #4,018, CC BY 4.0. Cheese from the USDA Economic Research Service; BlackRock opening price on the first trading day of each year from LSEG Analytics. Over 2002-2021; the differenced figures are calculated from the same data.

What both charts are missing

Ask the question this course always asks: what would one square be?

In both charts one square would be a year, not a person. There is no group of people the number could be about. Vigen notes on each page that using years as the base variable is itself the problem. Ask what one square would be, and there is nothing left to explain.

A confounder, not a cause

This is the structure that does real damage in medicine. It looks like cause and effect: a strong correlation, and the natural reading is that one drives the other. But a third factor sits behind the two and manufactures the link. Account for the confounder and the apparent cause disappears.

Statins prevent car crashes

141,086 people starting statins in British Columbia, grouped by how faithfully they took the pill. The faithful takers had fewer car crashes, workplace injuries, falls, fractures, burns, open wounds and poisonings. They also used more screening services, which a cholesterol pill has no way to affect.

Faithful takers versus not	Ratio
Hazard of motor vehicle accidents	0.75
Hazard of workplace accidents	0.77
Likelihood of using screening services	1.17

No mechanism in the pill reaches any of these. Something about the people produces all of them. The paper was published under the title "a cautionary tale".

Dormuth CR et al., Statin adherence and risk of accidents: a cautionary tale. Circulation 2009;119:2051-2057.

People who follow health instructions live longer

The same effect was visible thirty years earlier, inside a placebo arm. Five-year mortality was 15.1% among faithful takers of the dummy pill and 28.3% among unfaithful takers of the same dummy pill.

The pill was sugar. The difference was the person. The gap carries $P = 4.7 \times 10^{-16}$, significant at any bar you like, and still not the pill.

Coronary Drug Project Research Group, N Engl J Med 1980;303:1038-1041.

Bird watching does not prevent dementia

The headline is illustrative: no study makes this exact claim about birds. The real study is Whitehall II, 10,308 British civil servants followed for 28 years, measuring physical activity rather than birds. At first look, active people developed dementia less often.

Run the three explanations against it.

- **A coincidence?** No. These are people, and the grid can be drawn. One population, one split.
- **A common cause?** Maybe. Education, income, mobility, green space.
- **The illness itself?** Already under way, stopping the hobby years before anyone gives the disease a name.

Followed for the full 28 years, the activity made no difference: hazard ratio 1.00, confidence interval 0.80 to 1.24. Activity levels in people who would later be diagnosed were identical to everyone else's from 28 years before diagnosis down to about 10, and then fell away over the last nine. The correlation is real. The arrow is not.

Draw the network and three parents arrow into both bird watching and dementia: schooling and money, fixed by midlife; heart, joints and hearing, which develop late; and the illness already starting.

> Two of the three can be measured and adjusted for. The third is the illness itself, and no study can subtract it.

Note what the timing settles and what it doesn't. A late divergence fits the illness causing the withdrawal. It also fits vascular disease, frailty, hearing loss, depression or isolation, all of which develop late, all of which reduce getting out, and all of which raise dementia risk. Timing narrows the field. It does not by itself identify the arrow.

Sabia S et al., Physical activity, cognitive decline, and risk of dementia: 28 year follow-up of the Whitehall II cohort. BMJ 2017;357:j2709.

Smoking causes cancer: this time the arrow survives

The 1950s cohorts were enormous, and smokers were stricken far more often. R. A. Fisher, the era's leading statistician and the inventor of the significance test, objected: perhaps a gene produces both the craving and the cancer.

That is the middle card from earlier, a common cause, plus the deliberate claim that the arrow between smoking and cancer is not there at all. It was a legitimate question, and it is exactly the question this session teaches you to ask. What makes Fisher a cautionary tale is that he would accept no answer, and that he was a paid consultant to the tobacco industry while asking. He was also a smoker.

Fisher RA, Cigarettes, cancer, and statistics. Nature 1958;182:596. Stolley PD, Am J Epidemiol 1991;133:416-425.

One population, two denominators

Take 100 British men around 1950. About 65 smoked cigarettes. Of those 65, about 10 developed lung cancer by age 75. Of the other 35, about 1 did.

Direction	Calculation	Result
P(lung cancer | cigarettes)	10 of 65	16%, against 2% for never-smokers
P(cigarettes | lung cancer)	10 of 11	91%

The forward direction is the real finding: eight times apart. The reverse direction looks more dramatic and is worth much less, because it is inflated by the base rate. Two men in three smoked, so two in three of any group of men smoked, whether or not the group has anything to do with smoking. Two in three of the men run over by buses smoked too.

The forward comparison owes nothing to how common smoking was. That is the difference between the two directions, and it is Day 2's machinery applied to the one genuine cause in this session.

About 65% of British men smoked around 1950 (Doll R, Br J Cancer 1953;7:303). Lung cancer risk by age 75 is 15.7% for continuing smokers (Crispo A et al., Br J Cancer 2004;91:1280-1286); the 2% for never-smokers is approximate. Counts rounded to whole men.

The whole day, in four diagrams

Same dashed observation every time. Only the arrows differ.

- **Saturn and the degrees.** No arrow anywhere, and no population to draw, because the unit is a year.
- **The pill takers.** Two arrows out of something else. Dormuth's own conclusion was that adherers are systematically more health-seeking.
- **Bird watching.** The same arrow, pointing back.
- **Smoking.** The arrow survived.

> The data cannot pick the diagram for you. Ask which one the claim assumes, and what would rule the others out.

These are called causal diagrams. Reasoning with them by arithmetic, rather than by argument, is a Bayesian network. Pearl's The Book of Why is the entry point if you want to go further.

The five faults, and what to ask

1. **Relative reporting.** Ask: 36% of what? A change with no baseline is half a fact.
2. **Probability reversal.** A positive test is not a diagnosis. What it means depends on how common the condition is.
3. **Arbitrary categories.** Just over the line is still you. 119 to 120 moved the label, not your body.
4. **The Bernoulli fallacy.** One study is not proof. Run enough studies and something looks significant.
5. **The causal leap.** Moving together is not cause. Two things track, and a third often drives both.

Once you see them, you can't unsee them.

Course materials and contact

Slides, videos and the whitepaper library are at healthy-skepticism.com. Decks live in the student area, behind the password given out in class.

Murray Cantor PhD, murray@murraycantor.com

Joel Keenan MD, drkeenan@mdvip.com